

Project Demo Planning

Date	15 July 2026
Team ID	
Project Name	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction & Problem Statement	Introduce the team and present the problem: subjective oral grading, lack of acoustic metrics, no audit trail. Introduce SkillEcho as the solution.	3	Bhanu
2	Architecture Overview	Walk through the DDD architecture: Presentation (Streamlit) -> Application (FastAPI) -> AI (Whisper/SBERT/Librosa) -> Data (PostgreSQL). Show the src/ directory tree.	4	Sandeep
3	Technology Stack	Briefly highlight the 5 AI/ML technologies: Whisper (ASR), MiniLM Bi-Encoder, ms-marco Cross-Encoder, Librosa (DSP), ReportLab (PDF).	2	Sandeep
4	Live Demo: Registration & Login	Register a new student account on Streamlit. Log in and show the role-based dashboard.	2	Sandhya
5	Live Demo: Admin Concept Management	Log in as admin, create a new concept with a PDF upload, and show it appearing in the student view.	3	Sandhya
6	Live Demo: Full Audio Evaluation	Select a concept, record/upload a spoken explanation, submit, and display the scorecard with all metrics (similarity %, filler ratio, pause ratio, RMS, score, tier).	6	Sandeep
7	Live Demo: Topic Mismatch	Submit an off-topic audio clip; show the Topic Guardrail trigger giving a score of 0 and explain the compute savings from skipping heavy models.	2	Sandeep
8	Live Demo: PDF Report Download	Download the PDF report and walk through its sections: header, reference concept, transcript, waveform chart, metrics table, footer.	2	Sandhya
9	Live Demo: Evaluation History	Navigate to the history page and show past evaluations with scores and timestamps.	1	Sandhya
10	API Documentation	Open FastAPI /docs (Swagger UI) and show the 12 documented endpoints with request/response schemas.	1	Sandeep
11	Testing Summary	Summarise functional test coverage (30 features, 100% pass) and performance results (response time, RAM usage).	2	Archana & Tanuj
12	Q&A	Open floor for questions from the panel.	2	All

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	30-minute live demo on Streamlit Cloud (with localhost fallback); presents subjective grading, lack of acoustic metrics, and audit-trail gaps as the motivating problem.
2	Solution Overview	Covers the DDD architecture and the 5 core AI/ML technologies (Whisper, Bi-Encoder, Cross-Encoder, Librosa, ReportLab).
3	Live Feature Demonstration	Registration/login, admin concept management, full audio evaluation, topic mismatch handling, PDF report download, evaluation history, and API docs, each led by the responsible member.
4	Q&A Session	Open floor for the panel, backed by functional (100% pass, 30 features) and performance testing summaries.